Complex Equations

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Based on the lecture note uploaded on SPeCTRUM

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Learning Outcome

At the end of this lecture, you should be able to:

□ Describe and find the locus of a complex ~~number~~ equation in Cartesian plane.

Complex Equations

An equation with complex numbers as variables is called a complex equation.

Example:
Let z be a complex number, then
(i) $|z| = 2$; (ii) $\mathrm{Arg}(z) = \frac{\pi}{3}$; (iii)$\left|\frac{z+2i}{z+3-i}\right| = 2$; (iv) $|z-4i| + |z+3| = 2$
are complex equations.

By letting $z = x + iy$, a complex equation can be written in Cartesian form (in terms of x and y). $(x, y \in \mathbb{R})$

The set of all $z = x + iy$ satisfying a complex equation forms a curve in the xy-plane. This curve is called the locus of z.

Example

What is the locus of z satisfying $|z| = 2$?

Let $z = x + yi$

$|z| = 2 \Rightarrow |x+yi| = 2$

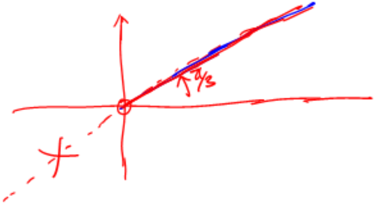
$\Rightarrow \sqrt{x^2+y^2} = 2$

$\Rightarrow x^2 + y^2 = 2^2$

The locus of z is a circle centred at $(0,0)$ with radius 2

Recall: $(x-a)^2 + (y-b)^2 = r^2$: circle centred at (a,b) with radius r.

What is the Cartesian form of the equation $\mathrm{Arg}(z) = \frac{\pi}{3}$ and what curve is represented by this equation?



$\mathrm{Arg}(z) = \pi/3$

Let $z = x + yi$, $x, y \in \mathbb{R}$.

$\tan \pi/3 = \frac{y}{x}$

$\Rightarrow \frac{y}{x} = \sqrt{3}$

$\Rightarrow y = \sqrt{3}x$, $x > 0$

The locus is a straight line consisting of points $(x, \sqrt{3}x)$ with $x > 0$.

(x, y)

Example

Describe the locus represented by

$$\left|\frac{z+2i}{z-i}\right| = 2$$

Let $z = x+yi$

$$\left|\frac{x+yi+2i}{x+yi-i}\right| = 2$$

$$\left|\frac{x+(y+2)i}{x+(y-1)i}\right| = 2$$

$$|x+(y+2)i| = 2\,|x+(y-1)i|$$

$$\sqrt{x^2+(y+2)^2} = 2\sqrt{x^2+(y-1)^2}$$

$$x^2 + (y+2)^2 = 4\left[x^2 + (y-1)^2\right]$$, squaring both sides of the equation.

$$x^2 + (y+2)^2 = 4x^2 + 4(y-1)^2$$

$$x^2 + y^2 + 4y + 4 = 4x^2 + 4y^2 - 8y + 4$$

$$3x^2 + 3y^2 - 12y = 0$$

$$x^2 + y^2 - 4y = 0$$

$$x^2 + (y-2)^2 - 4 = 0$$, completing the square.

$$x^2 + (y-2)^2 = 4 = 2^2$$

The locus is a circle centred at $(0,2)$ with radius 2.

$$\underline{x^2 + ax} = \underbrace{\left(x + \frac{a}{2}\right)^2} - \underline{\left(\frac{a}{2}\right)^2}$$

$$x^2 + ax + \left(\frac{a}{2}\right)^2$$

Example

If $|z| = 3$, what is the locus of the complex number $2z + 1$?

Let $w = 2z+1$ ← find the locus of w.

$$z = \frac{w-1}{2} = \frac{1}{2}(w-1)$$

$$|z| = 3 \Rightarrow \left|\frac{1}{2}(w-1)\right| = 3$$

$$\Rightarrow |w-1| = 6$$

Let $w = x+yi$

$$|x+yi-1| = 6 \Rightarrow |(x-1)+yi| = 6$$

$$\Rightarrow \sqrt{(x-1)^2+y^2} = 6$$

$$(x-1)^2 + y^2 = (6)^2$$

The locus is the circle centred at $C(1,0)$ with radius 6.

Example

Find the locus of z satisfying $|z - 2i| = 2\mathrm{Re}(z)$.

$|z-2i| = 2\mathrm{Re}(z) \leftarrow$

Let $z = x + yi$, $x, y \in \mathbb{R}$.

$|(x+yi)-2i| = 2x \leftarrow$ $x \geqslant 0$

$|x + (y-2)i| = 2x$

$\sqrt{x^2 + (y-2)^2} = 2x$

$x^2 + (y-2)^2 = 4x^2$

$(y-2)^2 = 3x^2$

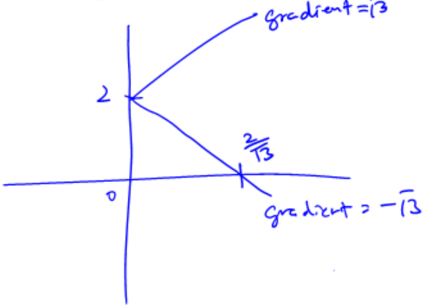
$|y-2| = |\sqrt{3}x| = \sqrt{3}|x|$

$|x| = \begin{cases} x, & x \geqslant 0 \\ -x, & x < 0 \end{cases}$

$$|y-2| = \sqrt{3}x$$

$$y-2 = \pm\sqrt{3}x$$

$$y = \underline{2+\sqrt{3}x}\,,\quad 2-\sqrt{3}x\,,\quad x \geq 0$$



Example

What is the locus of z satisfying $\mathrm{Arg}(z-1) = \frac{\pi}{4}$?

Let $z = x+yi$, $x, y \in \mathbb{R}$.

$\mathrm{Arg}(z-1) = \frac{\pi}{4}$ ← first quadrant.

$\mathrm{Arg}(x+yi-1) = \frac{\pi}{4}$

$\mathrm{Arg}((x-1)+yi) = \frac{\pi}{4}$

$\mathrm{Arg}(a+bi) = \frac{\pi}{4}$

$\tan\frac{\pi}{4} = \frac{y}{x-1}$

$x-1>0$

$x>1$

$\frac{y}{x-1} = 1$

$y = x-1, \quad x>1$